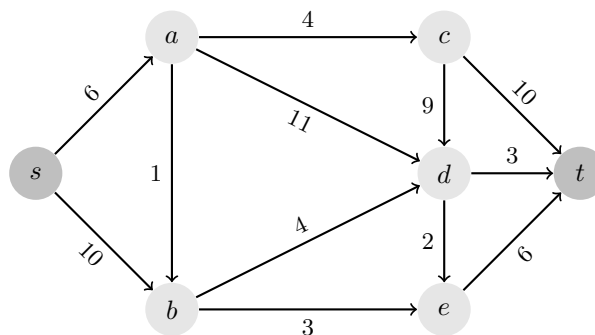


Algorithms and Probability**Peer Sheet P5****FS 2026**

The solutions for this sheet must be submitted on Moodle by **15 May, 15:59**. Afterwards you are assigned solutions of another student, which you have to grade (and upload to Moodle) until 11:59 of the following TUESDAY.

Exercise P5.1 – Flows and Cuts

Consider the following network N . The number on each edge represents its capacity.



- (a) Find a maximum flow f in N .
- (b) Construct the residual network (“Restnetzwerk”) corresponding to your flow f .
- (c) Find a minimum s - t -cut in N .
- (d) Prove that both your flow and your cut are indeed optimal.